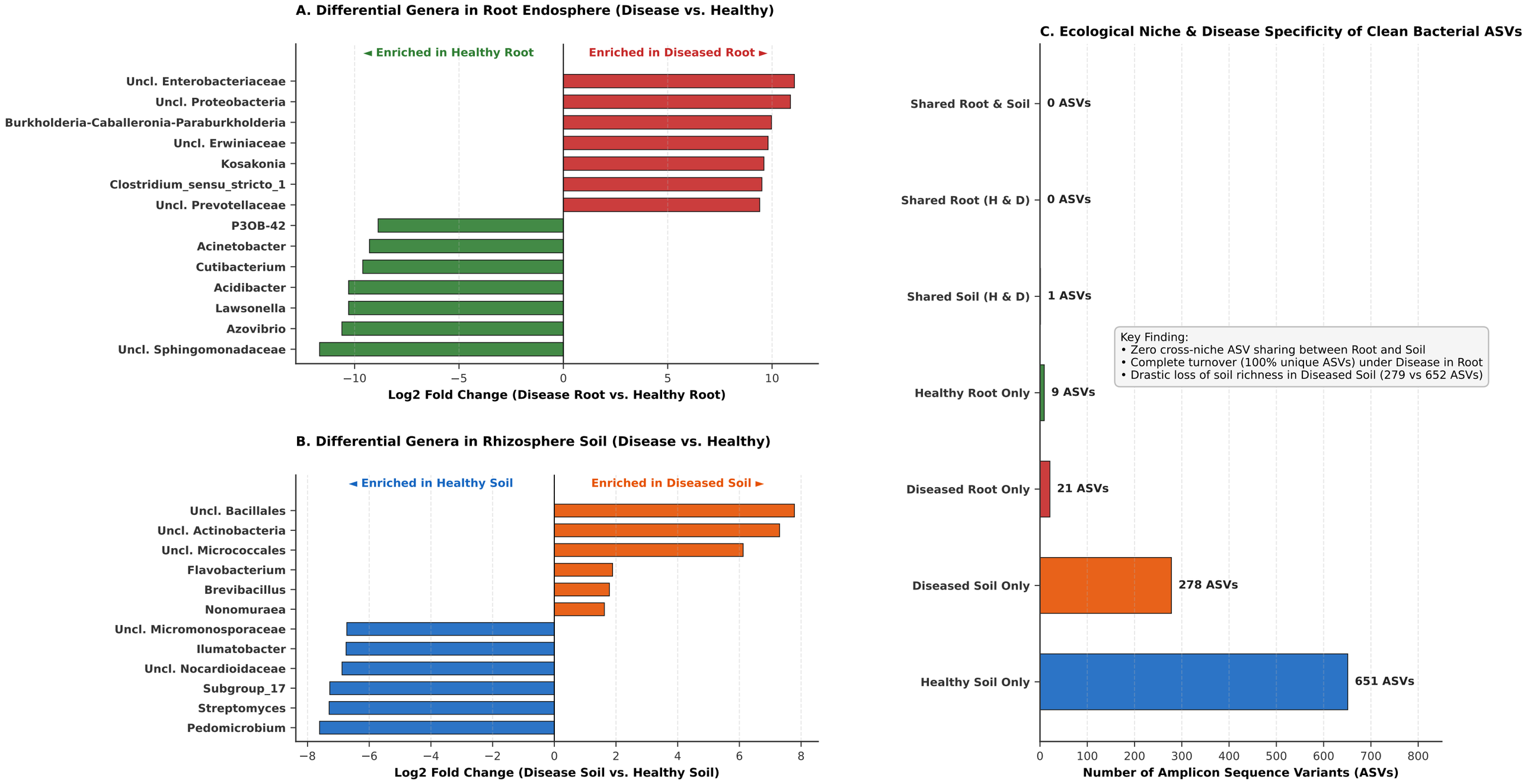
A. Differential Genera in Root Endosphere (Disease vs. Healthy)
◄ Enriched in Healthy Root
Enriched in Diseased Root ►
Uncl. Enterobacteriaceae
Uncl. Proteobacteria
Burkholderia-Caballeronia-Paraburkholderia
Uncl. Erwiniaceae
Kosakonia
Clostridium_sensu_stricto_1
Uncl. Prevotellaceae
P3OB-42
Acinetobacter
Cutibacterium
Acidibacter
Lawsonella
Azovibrio
Uncl. Sphingomonadaceae
−10
−5
0
5
10
Log2 Fold Change (Disease Root vs. Healthy Root)
B. Differential Genera in Rhizosphere Soil (Disease vs. Healthy)
◄ Enriched in Healthy Soil
Enriched in Diseased Soil ►
Uncl. Bacillales
Uncl. Actinobacteria
Uncl. Micrococcales
Flavobacterium
Brevibacillus
Nonomuraea
Uncl. Micromonosporaceae
Ilumatobacter
Uncl. Nocardioidaceae
Subgroup_17
Streptomyces
Pedomicrobium
−8
−6
−4
−2
0
2
4
6
8
Log2 Fold Change (Disease Soil vs. Healthy Soil)
C. Ecological Niche & Disease Specificity of Clean Bacterial ASVs
Shared Root & Soil
0 ASVs
Shared Root (H & D)
0 ASVs
Shared Soil (H & D)
1 ASVs
Key Finding:
• Zero cross-niche ASV sharing between Root and Soil
• Complete turnover (100% unique ASVs) under Disease in Root
• Drastic loss of soil richness in Diseased Soil (279 vs 652 ASVs)
Healthy Root Only
9 ASVs
Diseased Root Only
21 ASVs
Diseased Soil Only
278 ASVs
Healthy Soil Only
651 ASVs
0
100
200
300
400
500
600
700
800
Number of Amplicon Sequence Variants (ASVs)